NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

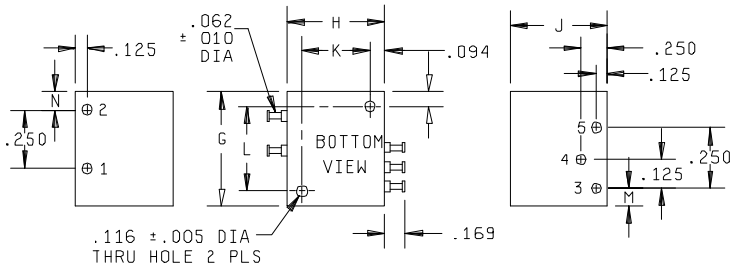
MIL-PRF-27/338D
22 January 1992
SUPERSEDING
MIL-T-27/338C
17 January 1986

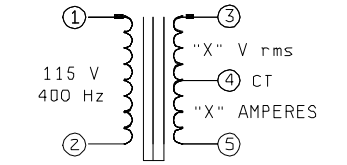
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 2 VOLT AMPERES, 400 HZ, TF5S03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





(SEE TABLE I FOR VALUE OF "X")
CIRCUIT DIAGRAM

Inches	mm
.005	0.13
.010	0.25
.031	0.79
.062	1.57
.094	2.39
.116	2.95
.125	3.18
.169	4.29
.250	6.35

Ltr	Dimensions	
	Inches	mm
G	.87 ±.015	22.1 ±0.38
H	.74 ±.015	18.8 ±0.38
J	.78 ±.015	19.8 ±0.38
K	.55 ±.015	14.0 ±0.38
L	.68 ±.015	17.3 ±0.38
M	.187 ±.031	4.75 ±0.79
N	.187 ±.031	4.75 ±0.79

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the top and sides of case.
4. Dimensional tolerance shall be ±.031 unless otherwise specified.
5. Dimensions G and H to be measured at top of transformer.

FIGURE 1. Dimensions and configuration.

(D) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIPEMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings

Primary voltage (1-2) 115 volts rms, 400 ±20 hertz

Secondary voltage and current (3-5). See table I

Voltampere 2 voltamperes

Working voltage (1-2) 535 volts peak

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Encapsulated

Material Diallyl phthalate

Terminals Turret type

Height 169 inch

Weight 06 pound, maximum

Operating temperature range -55°C to +130°C

Altitude 70,000 feet

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage (each winding)

At sea level 1500 volts rms

At reduced barometric pressure 1.25 times the peak working voltage specified

Electrical characteristics

Rated load With 115 volts rms and 400 Hz in (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I, ±5 percent

DC resistance (1-2) 201 ohms ±25 percent

Temperature rise 45°C with 115 volts rms, 400 hertz across (1-2) at an ambient temperature of 85°C, full load terminals (3-5)

$$\text{Regulation} \quad \frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$$ (Shall not exceed 10 percent)

Polarity Additive, with terminals 2 and 3 connected

Marking location See figure 1

Part or Identifying Number (PIN) M27/338- (dash number from table I)

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QUALITY ASSURANCE PROVISIONS

Extent of qualification

(D) Qualification testing and approval to M27/338-28 and M27/343-32 shall be sufficient to grant qualification approval to MIL-T-27/338 through MIL-T-27/343 inclusive, all parts

(D) Qualification testing and approval to M27/338-28 shall be sufficient to grant qualification approval to M27/338-01 through M27/338-28

TABLE I Electrical ratings

Dash number M27/338	Secondary voltage (3-5) (V rms) ±5% (CT)	Secondary current (3-5) (amperes)	Secondary DCR (3-5) (ohms) ±25%
-01	5 0	0 400	0 60
-02	6.3	0 317	1 06
-03	8.0	0.250	1 63
-04	10 0	0 200	2 50
-05	12 6	0 158	4 10
-06	14 0	0 142	5 59
-07	16.0	0 125	6 63
-08	18 0	0.111	9.00
-09	20.0	0 100	10 9

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TABLE I Electrical ratings - Continued

Dash number M27/338	Secondary voltage (3-5) (V rms) ±5% (CT)	Secondary current (3-5) (amperes)	Secondary DCR (3-5) (ohms) ±25%
-10	22 0	0 091	14.6
-11	25.0	0 080	20 4
-12	28 0	0 071	30 0
-13	31 0	0 064	25 6
-14	35 0	0 057	29 5
-15	39 0	0 051	38 6
-16	43 0	0 046	44 3
-17	48 0	0 041	57 0
-18	55 0	0 036	89 0
-19	63 0	0 031	100 0
-20	70 0	0 028	139 0
-21	80 0	0 025	163 0
-22	90 0	0 022	218 0
-23	100 0	0 020	283 0
-24	110 0	0 018	392 0
-25	123 0	0 016	495 0
-26	138 0	0 014	550 0
(D) -27	26 0	0 077	21 22
(D) -28	115 0	0 017	410 0

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CONCLUDING MATERIAL

Custodians
  Army - ER
  Navy - EC
  Air Force - 85

Review activities
  Army - AR
  Navy - OS
  Air Force - 11, 17, 99
  DLA - ES

User activities:
  Army - ME
  Navy - AS, MC
  Air Force - 19

Preparing activity:
  Army - ER

Agent
  DLA - ES

(Project 5950-0773)

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